Quasi-Poisson and negative binomial models

Last time: quasi-Poisson models

Poisson:

- Mean = λ_i
- Variance = λ_i

quasi-Poisson:

- Mean = λ_i
- Variance = $\phi\lambda_i$
- Variance is a linear function of the mean

Last time: Class activity

$(0.01646 - 0.01642) \pm 1.96\, SE(\hat{\beta}_3 - \hat{\beta}_4)$

```
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  2.8371917  0.0489922  57.911  < 2e-16 ***
male         0.4561700  0.0111074  41.069  < 2e-16 ***
age         -0.0068063  0.0006789 -10.026  < 2e-16 ***
education2   0.0164626  0.0126312   1.303 0.192462
education3   0.0164274  0.0160498   1.024 0.306057
education4  -0.0150328  0.0172291  -0.873 0.382921
diabetes    -0.0253978  0.0394181  -0.644 0.519368
BMI          0.0050011  0.0014065   3.556 0.000377 ***
```

Research question: Is there a difference in the average number of cigarettes smoked between smokers with some college education and smokers with a college degree, after accounting for sex, age, diabetes, and BMI?

> Can I calculate a confidence interval to answer this question, using just the R output shown here?

Class activity

Bootstrap resampling to compute the confidence interval:

- Take a sample with replacement from observed rows of the data
- Re-fit my (full) model on bootstrap sample
  - $\hat{\beta}_3^*$, $\hat{\beta}_4^*$
  - $\hat{\beta}_3^* - \hat{\beta}_4^*$
- Repeat many times (e.g. 1000, 10000, etc.)
- Compute $SD(\hat{\beta}_3^* - \hat{\beta}_4^*)$
- $(\hat{\beta}_3 - \hat{\beta}_4) \pm z^* \, SD(\hat{\beta}_3^* - \hat{\beta}_4^*)$

original data

from bootstrapping

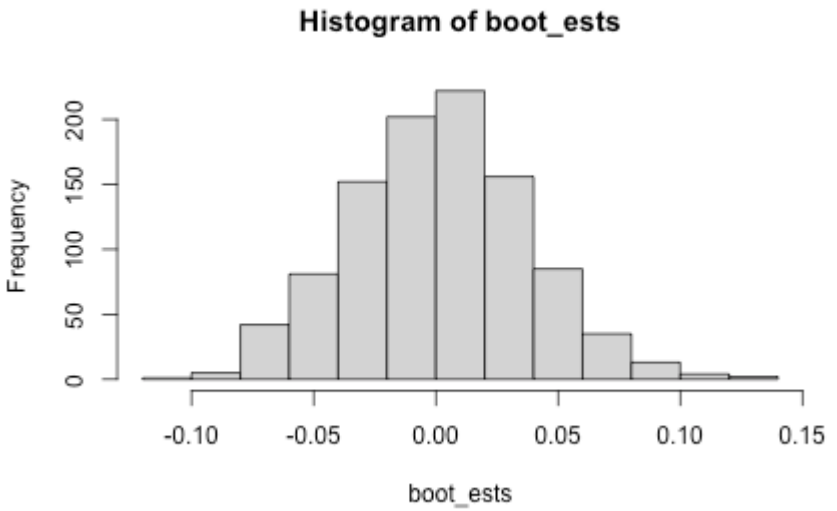
Class activity

95% CI: $(0.01646 - 0.01642) \pm 1.96(0.0362)$

$= (-0.071, 0.071)$

fail to reject $H_0: \beta_3 = \beta_4$

```
hist(boot_ests)
```




```
sd(boot_ests)
```

```
## [1] 0.03622715
```

Another question

```
m2 <- glm(cigsPerDay ~ male + age + education + diabetes + BMI,
          data = smokers, family = quasipoisson)
summary(m2)
```

```
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.837192   0.115099  24.650  < 2e-16 ***
male         0.456170   0.026095  17.481  < 2e-16 ***
age         -0.006806   0.001595  -4.267 2.07e-05 ***
education2   0.016463   0.029675   0.555    0.579
education3   0.016427   0.037706   0.436    0.663
education4  -0.015033   0.040477  -0.371    0.710
diabetes    -0.025398   0.092606  -0.274    0.784
BMI          0.005001   0.003304   1.513    0.130
```

Research question: Is there a relationship between education level and the number of cigarettes smoked per day, after accounting for sex, age, diabetes, and BMI?

$H_0: \beta_3 = \beta_4 = \beta_5 = 0$

$H_A:$ at least one of $\beta_3, \beta_4, \beta_5 \neq 0$

linear regression: $F = \frac{(SSE_{red} - SSE_{full}) / (df_{red} - df_{full})}{SSE_{full} / df_{full}}$ ← $\hat{\sigma}^2$

Nested tests with quasi-Poisson models

Research question: Is there a relationship between education level and the number of cigarettes smoked per day, after accounting for sex, age, diabetes, and BMI?

- In Poisson regression, we would use a likelihood ratio test
- However, the quasi-Poisson model includes the estimated dispersion, $\hat{\phi}$. We need to use an **F-test** instead

q = # parameters tested

$$F = \frac{(\text{deviance}_{red} - \text{deviance}_{full}) / (df_{red} - df_{full})}{\hat{\phi}_{full}}$$

$$\hat{\phi}_{full} = \frac{1}{df_{full}} \sum_{i=1}^{n} \hat{e}_{(P)i}^2$$

under H_0, $F \sim F_{q, n - p_{full}}$

numerator df → q; $n - p_{full}$ = df_{full} (denominator df)

Nested tests with quasi-Poisson models

```
m1 <- glm(cigsPerDay ~ male + age + education +
            diabetes + BMI,
          data = smokers, family = quasipoisson)
m2 <- glm(cigsPerDay ~ male + age + diabetes + BMI,
          data = smokers, family = quasipoisson)

m1$deviance
```

```
## [1] 11539.56
```

$$F = \frac{(11543.84 - 11539.56)/(2007 - 2004)}{5.52}$$

```
m2$deviance
```

```
## [1] 11543.84
```

under H_0, $F \sim F_{3,2004}$

```
m1$df.residual
```

```
## [1] 2004
```

```
m2$df.residual
```

```
## [1] 2007
```

Nested tests with quasi-Poisson models

```
m1$deviance
```

```
## [1] 11539.56
```

```
m2$deviance
```

```
## [1] 11543.84
```

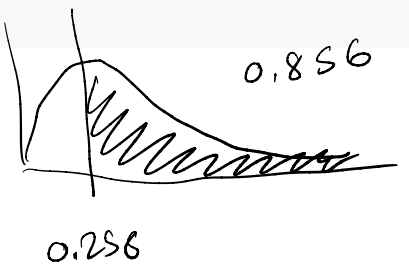
"probabilities for F distribution"

F

look in right tail

```
pf(0.258, 3, 2004, lower.tail=F)
```

```
## [1] 0.8556648
```



(fail to reject H_0: $\beta_3 = \beta_4 = \beta_5 = 0$)

Limitations of quasi-Poisson models

Important: quasi-Poisson models do *not* describe a full distribution for the response (there is no "quasi-Poisson" distribution). They describe *only* the mean and variance of the response.

As a result, one cannot calculate probabilities for a quasi-Poisson:

```
m2 <- glm(cigsPerDay ~ male + age + diabetes + BMI,
          data = smokers, family = quasipoisson)
rootogram(m2)
```

```
Error in prodist.glm(object) :
family is not associated with a full probability distributi
```

Comparing Poisson and quasi-Poisson

Poisson:

- Mean = λ_i
- Variance = λ_i

quasi-Poisson:

- Mean = λ_i
- Variance = $\phi\lambda_i$
- Variance is a linear function of the mean

> What if we want variance to depend on the mean in a different way?

Introducing the negative binomial

If $Y_i \sim NB(\theta, p)$, then Y_i takes values $y = 0, 1, 2, 3, \ldots$ with probabilities

$$P(Y_i = y) = \frac{(y + \theta - 1)!}{y!(\theta - 1)!}(1 - p)^\theta p^y$$

- $\theta > 0, \quad p \in [0, 1]$
- Mean = $\frac{p\theta}{1 - p} = \mu$
- Variance = $\frac{p\theta}{(1 - p)^2} = \mu + \frac{\mu^2}{\theta}$
- Variance is a *quadratic* function of the mean

Mean and variance for a negative binomial variable

If we keep μ fixed and $\theta \to \infty$, then $NB \to Poisson(\mu)$

If $Y_i \sim NB(\theta, p)$, then

- Mean = $\frac{p\theta}{1-p} = \mu$
- Variance = $\frac{p\theta}{(1-p)^2} = \mu + \frac{\mu^2}{\theta}$

> How is θ related to overdispersion?

θ small (μ fixed): $\mu + \frac{\mu^2}{\theta} >> \mu$ (more overdispersion from Poisson perspective)

θ large (μ fixed): $\mu + \frac{\mu^2}{\theta} \approx \mu$

Negative binomial regression

$$Y_i \sim NB(\theta,\ p_i)$$

$$\log(\mu_i) = \beta_0 + \beta_1 X_i$$

- $\mu_i = \dfrac{p_i \theta}{1 - p_i}$
- Note that θ is the same for all i
- Note that just like in Poisson regression, we model the average count
  - Interpretation of βs is the same as in Poisson regression

Comparing Poisson, quasi-Poisson, negative binomial

Poisson:

- Mean = λ_i
- Variance = λ_i

quasi-Poisson:

- Mean = λ_i
- Variance = $\phi\lambda_i$

negative binomial:

- Mean = μ_i
- Variance = $\mu_i + \frac{\mu_i^2}{\theta}$

In R

don't have to specify a family

```
library(MASS)
m3 <- glm.nb(cigsPerDay ~ male + age + education +
               diabetes + BMI, data = smokers)
```

```
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  2.877771   0.123477  23.306  < 2e-16 ***
male         0.459148   0.027641  16.611  < 2e-16 ***
age         -0.007010   0.001731  -4.050 5.12e-05 ***
education2   0.024518   0.032534   0.754    0.451
education3   0.009252   0.040802   0.227    0.821
education4  -0.027732   0.044825  -0.619    0.536
diabetes    -0.010124   0.099126  -0.102    0.919
BMI          0.003693   0.003573   1.033    0.301

(Dispersion parameter for Negative Binomial(3.2981) family
```

($\alpha = 1$)

$\hat{\theta}$

$\hat{\theta} = 3.30$

In R

$\exp(-0.007) \approx 0.993$

	Estimate	Std. Error	z value	Pr(>|z|)	
(Intercept)	2.877771	0.123477	23.306	< 2e-16	***
male	0.459148	0.027641	16.611	< 2e-16	***
age	-0.007010	0.001731	-4.050	5.12e-05	***
education2	0.024518	0.032534	0.754	0.451	
education3	0.009252	0.040802	0.227	0.821	
education4	-0.027732	0.044825	-0.619	0.536	
diabetes	-0.010124	0.099126	-0.102	0.919	
BMI	0.003693	0.003573	1.033	0.301	

> How do I interpret the estimated coefficient -0.007?

A one-year increase in age is associated with a 0.7% decrease in mean # of cigarettes smoked, holding other variables constant

Hypothesis testing

```
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  2.877771   0.123477  23.306  < 2e-16 ***
male         0.459148   0.027641  16.611  < 2e-16 ***
age         -0.007010   0.001731  -4.050 5.12e-05 ***
education2   0.024518   0.032534   0.754    0.451
education3   0.009252   0.040802   0.227    0.821
education4  -0.027732   0.044825  -0.619    0.536
diabetes    -0.010124   0.099126  -0.102    0.919
BMI          0.003693   0.003573   1.033    0.301
```

Research question: Is there a relationship between education level and the number of cigarettes smoked per day, after accounting for sex, age, diabetes, and BMI?

Poisson (or logistic):

$$G = \text{deviance}_{\text{reduce}} - \text{deviance}_{\text{full}}$$

$$= 2\log L_{\text{full}} - 2\log L_{\text{reduced}}$$

$$= 2\log\left(\frac{L_{\text{full}}}{L_{\text{reduced}}}\right)$$

(Likelihood ratio test)

NB: we can't use deviance directly,

but can still compute

$$G = 2\log L_{\text{full}} - 2\log L_{\text{reduced}}$$

Likelihood ratio test

```
m3 <- glm.nb(cigsPerDay ~ male + age + education +
               diabetes + BMI, data = smokers)
m4 <- glm.nb(cigsPerDay ~ male + age +
               diabetes + BMI, data = smokers)
m3$twologlik - m4$twologlik
```

```
## [1] 1.423055
```

```
pchisq(1.423, df=3, lower.tail=F)
```

```
## [1] 0.7001524
```

Under H_0: $G \sim \chi^2_q$ q = # parameters tested

(Alternative: parametric bootstrapping)

quasi-Poisson vs. negative binomial

quasi-Poisson:

- linear relationship between mean and variance
- easy to interpret $\widehat{\phi}$
- same as Poisson regression when $\phi = 1$
- simple adjustment to estimated standard errors
- estimated coefficients same as in Poisson regression

negative binomial:

- quadratic relationship between mean and variance
- we get to use a likelihood, rather than a quasi-likelihood
- Same as Poisson regression when θ is very large and p is very small

Class activity

Work on the class activity. Submit the rendered HTML on Canvas at the end of class.

Class activity

```
m_art <- glm.nb(art ~ phd + ment + kid5, data = artic
```

```
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  0.233528   0.117585   1.986   0.0470 *
phd          0.008203   0.036069   0.227   0.8201
ment         0.030374   0.003240   9.373   <2e-16 ***
kid5        -0.093256   0.046475  -2.007   0.0448 *
```

Class activity

```
m_art <- glm.nb(art ~ phd + ment + kid5, data = artic
m_art_reduced <- glm.nb(art ~ phd + kid5,
                        data = articles)
m_art$twologlik - m_art_reduced$twologlik
```

```
## [1] 77.00159
```

```
pchisq(77, df=1, lower.tail=F)
```

```
## [1] 1.70958e-18
```